MID SEMESTER TEST - 1

B.TECH. - I Year/II Semester (Branches: EI, IP, IT & ME)

PH10009: APPLIED PHYSICS (THEORY)

Time : 1 hour Date: Feb. 12, 2026 Max Marks : 20

Name:_____ **Roll Number: 25**_____ **Student Signature:**_____

Signature of Invigilator:_____ **Signature of Examiner:**_____

All questions are compulsory. No extra sheets will be given. Use margin space for rough work.

S.No.	Question	Mark	CO	BL
Q1	A fiber having n_{co}=1.46 and n_{cl}=1.455 is coupled with He-Ne laser. Estimate the maximum core radius so that it becomes single mode fiber for maximum wavelength and multimode for minimum.	3	1,2	2
Q2	Draw Energy level diagram of He-Ne and Ruby laser write every energy level and transition wavelength in terms of eV.	2	1,2	4

Ruby Laser **He-Ne Laser**

Q3	Let us consider that the end of an optical fiber is not perpendicular to the optical axis while it is cleaved at an angle α. Drive the Expression for the Numerical aperture of fiber.	3	1	1

Q4	Mention the major contribution (Device/Phenomenon/Invention) of following scientist: (a) C.K.N. Patel =_____ _____ (b) N.S. Kapany =_____ _____ (c) M.L. Bhaumik =_____ _____ (d) S.N. Bose =_____ _____	2	1,2	6
Q5	Let the losses and length of the cavity of 3-level and 4-level lasers are same. If the pumping rate of both the laser is doubled, which one show greater percentage increase in output power and why?	2	1	4
Q6	Write **TRUE/FALSE**, justify your answer. (a) None of the fiber is a single mode fiber (b) Two level laser is not possible (c) Length of the cavity is highly significant for selective lasing (d) Population inversion is a negative temperature state	6	2	3
Q7	Write the expression for the ratio of number of spontaneous to stimulated emission and calculate its value by considering the optical frequency of visible light source kept at 6000K.	2	2	3

Rough Work

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S.No.	Question / Answer

Q1

$$\text{NA} = \sqrt{n_{co}^2 - n_{cl}^2} = \sqrt{(1.46)^2 - (1.455)^2} \approx 0.121$$

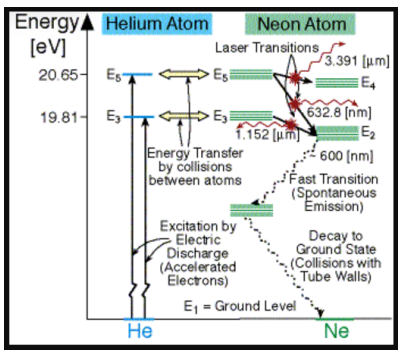
For single-mode operation $V \leq 2.405$. This value must be obtained at $\lambda_{\max} = 3.39\,\mu$m,

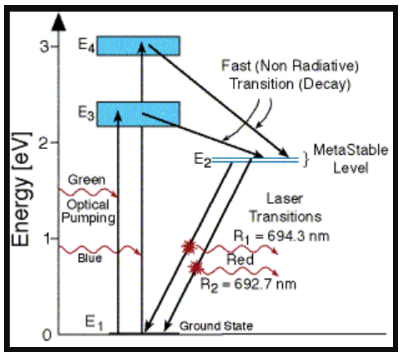
$$V = \frac{2\pi a}{\lambda} \times NA \Rightarrow 2.405 = \frac{2\pi a}{3.39} \times 0.121 \Rightarrow \boxed{\textbf{Core Radius}, a \approx \textbf{10.7}\,\mu\text{m}}$$

$$\text{At } \lambda_{\min} = 0.6328\,\mu\text{m}, V = \frac{2\pi(10.7)}{0.6328} \times 0.121 \approx 12.9$$

Since $V > 2.405$, the fiber is multimode at minimum wavelength.

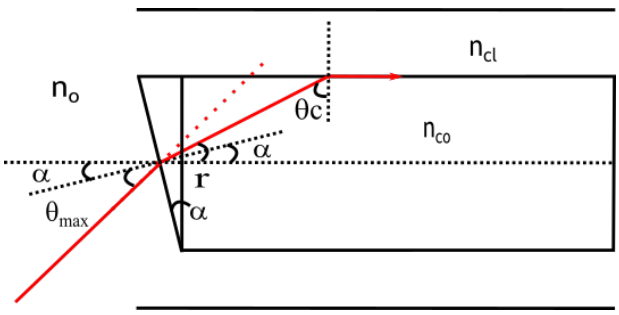
Q2





Laser	Transition	Wave-length	Ener. (eV)
He–Ne	$3s_2 \to 2p_4$	632.8 nm	1.96
He–Ne	$3s_2 \to 3p_4$	$3.39\,\mu$m	0.366
He–Ne	$3s_1 \to 3p_4$	$1.152\,\mu$m	1.08
Ruby	${}^2E \to {}^4A_2$	694.3 nm	1.785
Ruby	${}^2E \to {}^4A_2$	692.7 nm	1.790

Q3



$$n_0 \sin(\theta + \alpha) = n_{co} \sin r, \quad \sin\theta_c = \frac{n_{cl}}{n_{co}}$$

$$\sin r = \sqrt{1 - \left(\frac{n_{cl}}{n_{co}}\right)^2}$$

$$\boxed{\text{NA} = n_0 \sin(\theta_{max} + \alpha) = \sqrt{n_{co}^2 - n_{cl}^2}}$$

Q4 (a) C.K.N. Patel – Invented CO_2 laser (1964).
(b) N.S. Kapany – Father of fiber optics (1950).
(c) M.L. Bhaumik – Excimer laser development (1973).
(d) S.N. Bose – Bose–Einstein statistics (1924).

Q5 **Ans:** A 4-level laser shows a greater percentage increase in output power because of its low threshold pumping rate.

Q6 (a) TRUE — $V < 2.405$ for single-mode; since $V \propto 1/\lambda$, a fiber can behave as single or multi mode.
(b) TRUE — A two-level laser cannot achieve population inversion.
(c) TRUE — Cavity length controls mode spacing and frequency selectivity.
(d) TRUE —

$$\frac{N_2}{N_1} = \exp\left(-\frac{E_2 - E_1}{kT}\right) \Rightarrow T < 0$$

Population inversion corresponds to negative temperature.

Q7

$$\frac{R_{sp}}{R_{st}} = \frac{A_{21}}{B_{21}\rho(\nu)} = e^{\frac{h\nu}{kT}} - 1 \text{ At } T = 6000\,\text{K}, \nu = 5 \times 10^{14}\,\text{Hz} : \frac{h\nu}{kT} \approx 4 \Rightarrow \frac{R_{sp}}{R_{st}} \approx 54$$